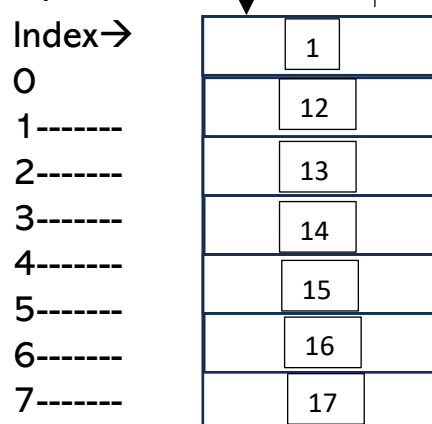


Collection in Java

- Collections API is used to collect group of Objects.
- java.util package is used to implement collection.
- It implements many data structure like Set, List, Stack, Queue etc .. in Java.
- <E> represents Generic Type.it can be anything like String,Integer Float, Double etc (We pass Wrapper class in E).
 - **int** → Primitive data type
 - **Integer** → Wrapper Class
 - **float** → Primitive data type
 - **Float** → Wrapper class
 - **double** → Primitive data type
 - **Double** → Wrapper class

- In Stack ,index 0 start from right going in -ve from right to left i.e.

0 → 1 → 2 → 3 → -4



- First In Last Out
- `javap java.util.Stack`
- `public boolean empty();` // check if stack is empty or not
 - [12, 13, 13, 16, 17]
- Index: 5 4 3 2 1

Queue Implementation:

- First In First Out (FIFO)
- `javap java.util.Queue`

- To Traverse in List, Stack or Queue or any data structure we use iterator Interface (to visit items inside these).
- java.util.Iterator
 - hasNext → condition
 - next() → Access the item
- Iterator VS iterator : One is Interface another is method (smallcase).

Set:

- No Duplicate Entries allowed
- javap java.util.Set

Linked List Implementation:

- class
- javap java.util.LinkedList

Sorting in Collection(M.imp):

- Collection → Interface
- Collections → Class\

Sorting through Comparable Interface (M.imp):

- java.lang.comparable
- javap java.util.Comparator
- E : element
- T : Type